

labsheet=09 PRIYANSHU SHARMA CU240250980 SECTION(B) (47) Mastery in Data Analytics and Visualizations and Career Advancement (Technical)

```
# EXPERIMENT NO. 9
# DATA STORYTELLING AND BUSINESS INSIGHT GENERATION
# USING INTERACTIVE VISUALIZATIONS
# =====

# -----
# 1. IMPORT LIBRARIES
# -----

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

print("=" * 70)
print("DATA STORYTELLING AND BUSINESS INSIGHT GENERATION")
print("=" * 70)
print("priyanshu sharma cu240250980")
```

```
=====
DATA STORYTELLING AND BUSINESS INSIGHT GENERATION
=====
priyanshu sharma cu240250980
```

```
# =====
# 1. IMPORT LIBRARIES
# =====

import pandas as pd

# =====
# 2. LOAD DATASET
# =====

file_path = "superstore.csv"

try:
    df = pd.read_csv(file_path, encoding="latin1")

except FileNotFoundError:
    print("\nERROR: Dataset file not found!")
    print("Make sure 'superstore.csv' is in the same folder as this Python file.")
    exit()

print("\nDataset loaded successfully!")
print("priyanshu sharma cu240250980")

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 5 Rows:")
print(df.head())
```

Dataset loaded successfully!
priyanshu sharma cu240250980

Dataset Shape:
(51290, 27)

First 5 Rows:

	Category	City	Country	Customer.ID	Customer.Name \
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert
3	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert
4	Office Supplies	Los Angeles	United States	AP-109154	Arthur Pritchep

	Discount	Market	Order.Date	Order.ID	...	\
0	0.0	US	1 2011-01-07 00:00:00.000	CA-2011-130813	...	
1	0.0	US	1 2011-01-21 00:00:00.000	CA-2011-148614	...	
2	0.0	US	1 2011-08-05 00:00:00.000	CA-2011-118962	...	
3	0.0	US	1 2011-08-05 00:00:00.000	CA-2011-118962	...	
4	0.0	US	1 2011-09-29 00:00:00.000	CA-2011-146969	...	

	Sales	Segment	Ship.Date	Ship.Mode	Shipping.Cost	\
0	19	Consumer	2011-01-09 00:00:00.000	Second Class	4.37	
1	19	Consumer	2011-01-26 00:00:00.000	Standard Class	0.94	
2	21	Consumer	2011-08-09 00:00:00.000	Standard Class	1.81	
3	111	Consumer	2011-08-09 00:00:00.000	Standard Class	4.59	
4	6	Consumer	2011-10-03 00:00:00.000	Standard Class	1.32	

	State	Sub.Category	Year	Market2	weeknum
0	California	Paper	2011	North America	2
1	California	Paper	2011	North America	4
2	California	Paper	2011	North America	32
3	California	Paper	2011	North America	32
4	California	Paper	2011	North America	40

[5 rows x 27 columns]

3. DISPLAY BASIC INFORMATION

```
print("\n=====")
print("DATASET INFORMATION")
print("=====")
```

```
print("\nFirst 5 Records:")
print(df.head())
```

```
print("\nDataset Shape:")
print(df.shape)
```

```
print("\nColumn Names:")
print(df.columns.tolist())
```

```
print("\nData Types:")
print(df.dtypes)
```

```
print("\nMissing Values:")
print(df.isnull().sum())
print("priyanshu sharma cu240250980")
```

```
Country      object
Customer.ID   object
```

```
Ship.Date      0
Ship.Mode      0
Shipping.Cost   0
State          0
Sub.Category    0
Year           0
Market2        0
weeknum        0
dtype: int64
priyanshu sharma cu240250980
```

```
# 4. CLEAN COLUMN NAMES
```

```
# -----
```

```
df.columns = df.columns.str.strip()

print("\nCleaned Column Names:")
print(df.columns.tolist())
print("priyanshu sharma cu240250980")
```

```
Cleaned Column Names:
```

```
['Category', 'City', 'Country', 'Customer.ID', 'Customer.Name', 'Discount', 'Market', 'è°°â½\x95æ\x95°', 'Order.Date', 'Order']
priyanshu sharma cu240250980
```

```
# 5. CONVERT DATE COLUMN
```

```
# -----
```

```
if "Order Date" in df.columns:
    df["Order Date"] = pd.to_datetime(
        df["Order Date"],
        errors="coerce"
    )

if "Ship Date" in df.columns:
    df["Ship Date"] = pd.to_datetime(
        df["Ship Date"],
        errors="coerce"
    )
```

```
# 6. REMOVE DUPLICATE RECORDS
```

```
# -----
```

```
duplicate_count = df.duplicated().sum()

print("\nNumber of Duplicate Records:", duplicate_count)

df = df.drop_duplicates()

print("Duplicates removed.")
print("New Dataset Shape:", df.shape)
```

```
Number of Duplicate Records: 0
```

```
Duplicates removed.
```

```
New Dataset Shape: (51290, 27)
```

```
# 7. BASIC STATISTICAL ANALYSIS
```

```
# -----
```

```
print("\n=====")
print("STATISTICAL SUMMARY")
print("=====")

print(df.describe(include="all"))
print("priyanshu sharma cu240250980")
```

```
50%      NaN      NaN      NaN      NaN
75%      NaN      NaN      NaN      NaN
```

```

max      NaN      0.850000      NaN      1.0

      Order.Date      Order.ID      ...      Sales      Segment \
count      51290      51290      ...      51290.000000      51290
unique      1430      25035      ...      NaN      3
top      2014-06-18 00:00:00.000      CA-2014-100111      ...      NaN      Consumer
freq      135      14      ...      NaN      26518
mean      NaN      NaN      ...      246.498440      NaN
std      NaN      NaN      ...      487.567175      NaN
min      NaN      NaN      ...      0.000000      NaN
25%      NaN      NaN      ...      31.000000      NaN
50%      NaN      NaN      ...      85.000000      NaN
75%      NaN      NaN      ...      251.000000      NaN
max      NaN      NaN      ...      22638.000000      NaN

      Ship.Date      Ship.Mode      Shipping.Cost      State \
count      51290      51290      51290.000000      51290
unique      1464      4      NaN      1094
top      2014-11-22 00:00:00.000      Standard Class      NaN      California
freq      130      30775      NaN      2001
mean      NaN      NaN      NaN      26.375818      NaN
std      NaN      NaN      NaN      57.296810      NaN
min      NaN      NaN      NaN      0.002000      NaN
25%      NaN      NaN      NaN      2.610000      NaN
50%      NaN      NaN      NaN      7.790000      NaN
75%      NaN      NaN      NaN      24.450000      NaN
max      NaN      NaN      NaN      933.570000      NaN

      Sub.Category      Year      Market2      weeknum
count      51290      51290.000000      51290      51290.000000
unique      17      NaN      6      NaN
top      Binders      NaN      APAC      NaN
freq      6152      NaN      11002      NaN
mean      NaN      2012.777208      NaN      31.287112
std      NaN      1.098931      NaN      14.429795
min      NaN      2011.000000      NaN      1.000000
25%      NaN      2012.000000      NaN      20.000000
50%      NaN      2013.000000      NaN      33.000000
75%      NaN      2014.000000      NaN      44.000000
max      NaN      2014.000000      NaN      53.000000

```

[11 rows x 27 columns]

priyanshu.sharma_cu240250080

8. KEY PERFORMANCE INDICATORS (KPIs)

```

# -----

print("\n=====")
print("KEY PERFORMANCE INDICATORS")
print("=====")

if "Sales" in df.columns:
    total_sales = df["Sales"].sum()
    print(f"Total Sales      : ${total_sales:,.2f}")

if "Profit" in df.columns:
    total_profit = df["Profit"].sum()
    print(f"Total Profit       : ${total_profit:,.2f}")

if "Quantity" in df.columns:
    total_quantity = df["Quantity"].sum()
    print(f"Total Quantity      : {total_quantity:,.0f}")

if "Customer ID" in df.columns:
    total_customers = df["Customer ID"].nunique()
    print(f"Total Customers     : {total_customers:,}")

if "Order ID" in df.columns:
    total_orders = df["Order ID"].nunique()
    print(f"Total Orders        : {total_orders:,}")

```

```

=====
KEY PERFORMANCE INDICATORS
=====
Total Sales      : $12,642,905.00
Total Profit     : $1,467,457.29
Total Quantity   : 178,312

```

9. PROFIT MARGIN

```

# -----

if "Sales" in df.columns and "Profit" in df.columns:

    profit_margin = (

```

```

        df["Profit"].sum() /
        df["Sales"].sum()
    ) * 100

print(f"Profit Margin      : {profit_margin:.2f}%")
print("priyanshu sharma cu240250980")

```

```

Profit Margin      : 11.61%
priyanshu sharma cu240250980

```

```

# =====
# IMPORT LIBRARIES
# =====

import pandas as pd
import matplotlib.pyplot as plt

# =====
# DATA STORY 1
# SALES BY CATEGORY
# =====

# Check required columns
print("Available columns:")
print(df.columns.tolist())

if "Category" in df.columns and "Sales" in df.columns:

    # Convert Sales to numeric
    df["Sales"] = pd.to_numeric(
        df["Sales"],
        errors="coerce"
    )

    # Remove missing Sales values
    df = df.dropna(subset=["Sales"])

    # Calculate sales by category
    category_sales = (
        df.groupby("Category")["Sales"]
        .sum()
        .sort_values(ascending=False)
    )

    print("\n=====")
    print("SALES BY CATEGORY")
    print("=====")

    print(category_sales)

    # Create bar chart
    plt.figure(figsize=(10, 6))

    category_sales.plot(
        kind="bar"
    )

    plt.title("Sales by Category")
    plt.xlabel("Category")
    plt.ylabel("Total Sales")

    plt.xticks(rotation=0)

    plt.tight_layout()
    plt.show()

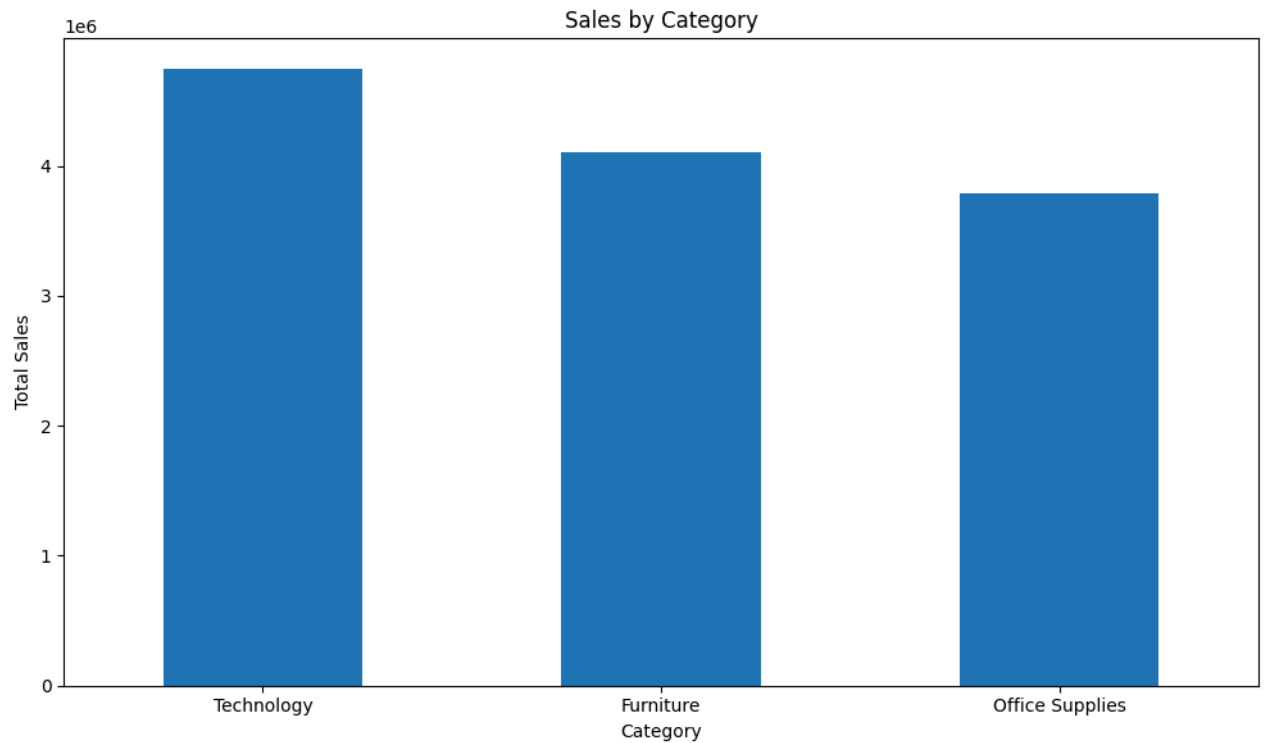
else:
    print("\nERROR: 'Category' or 'Sales' column not found!")

```

Available columns:
 ['Category', 'City', 'Country', 'Customer.ID', 'Customer.Name', 'Discount', 'Market', 'è°ä%x95æ\x95°', 'Order.Date', 'Order

SALES BY CATEGORY

```
Category
Technology      4744691
Furniture       4110884
Office Supplies  3787330
Name: Sales, dtype: int64
```



```
# DATA STORY 2
# PROFIT BY CATEGORY
# =====

if "Category" in df.columns and "Profit" in df.columns:

    category_profit = (
        df.groupby("Category")["Profit"]
        .sum()
        .sort_values(ascending=False)
    )

    print("\n=====")
    print("PROFIT BY CATEGORY")
    print("=====")
    print("priyanshu sharma cu240250980")
    print(category_profit)

    plt.figure(figsize=(10, 6))

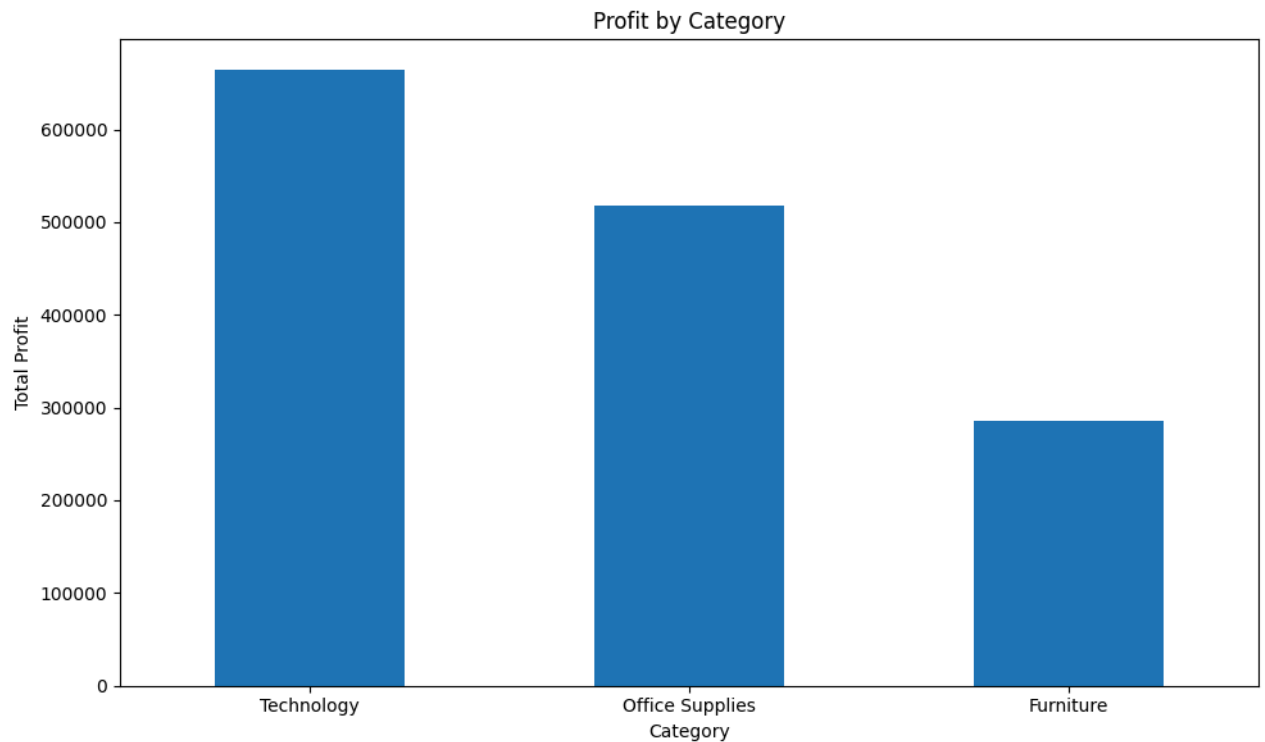
    category_profit.plot(
        kind="bar"
    )

    plt.title("Profit by Category")
    plt.xlabel("Category")
    plt.ylabel("Total Profit")

    plt.xticks(rotation=0)

    plt.tight_layout()
    plt.show()
```

```
=====
PROFIT BY CATEGORY
=====
priyanshu sharma cu240250980
Category
Technology      663778.73318
Office Supplies  518473.83430
Furniture        285204.72380
Name: Profit, dtype: float64
```



```
# DATA STORY 3
# SALES TREND OVER TIME
# =====

if "Order Date" in df.columns and "Sales" in df.columns:

    monthly_sales = (
        df.set_index("Order Date")
        .resample("ME")["Sales"]
        .sum()
    )

    print("\n=====")
    print("MONTHLY SALES TREND")
    print("=====")
    print("priyanshu sharma cu240250980")
    print(monthly_sales.head(12))

    plt.figure(figsize=(14, 6))

    plt.plot(
        monthly_sales.index,
        monthly_sales.values,
        marker="o"
    )

    plt.title("Monthly Sales Trend")
    plt.xlabel("Month")
    plt.ylabel("Sales")

    plt.xticks(rotation=45)

    plt.tight_layout()
    plt.show()
```

```
# DATA STORY 4
# SALES BY REGION
# =====

if "Region" in df.columns and "Sales" in df.columns:

    region_sales = (
        df.groupby("Region")["Sales"]
        .sum()
        .sort_values(ascending=False)
    )

    print("\n=====")
    print("SALES BY REGION")
    print("=====")
    print("priyanshu sharma cu240250980")
    print(region_sales)

    plt.figure(figsize=(10, 6))

    region_sales.plot(
        kind="bar"
    )

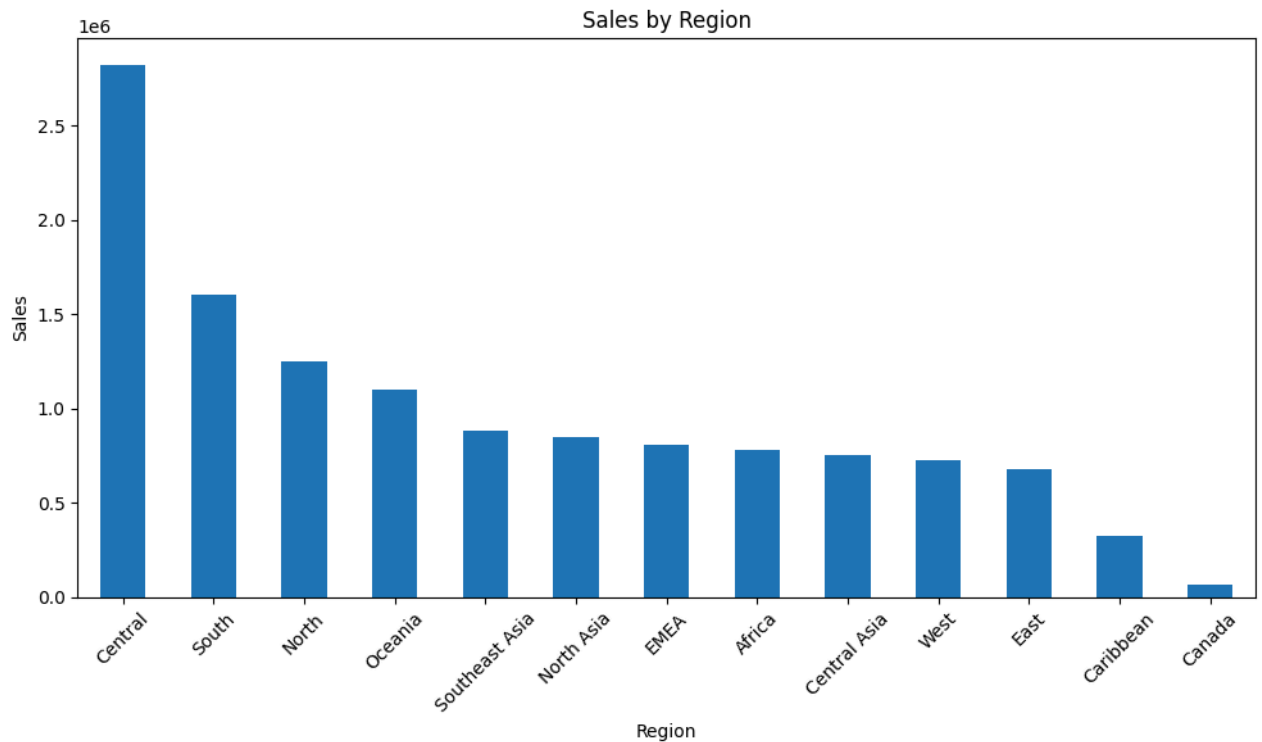
    plt.title("Sales by Region")
    plt.xlabel("Region")
    plt.ylabel("Sales")

    plt.xticks(rotation=45)

    plt.tight_layout()
    plt.show()
```



```
=====
SALES BY REGION
=====
priyanshu sharma cu240250980
Region
Central      2822399
South        1600960
North        1248192
Oceania       1100207
Southeast Asia  884438
North Asia    848349
EMEA          806184
Africa        783776
Central Asia   752839
West          725514
East          678834
Caribbean     324281
Canada        66932
Name: Sales, dtype: int64
```



```
# DATA STORY 5
# PROFIT BY REGION
# =====

if "Region" in df.columns and "Profit" in df.columns:

    region_profit = (
        df.groupby("Region")["Profit"]
        .sum()
        .sort_values(ascending=False)
    )

    print("\n=====")
    print("PROFIT BY REGION")
    print("=====")
    print("priyanshu sharma cu240250980")
    print(region_profit)

    plt.figure(figsize=(10, 6))

    region_profit.plot(
        kind="bar"
    )

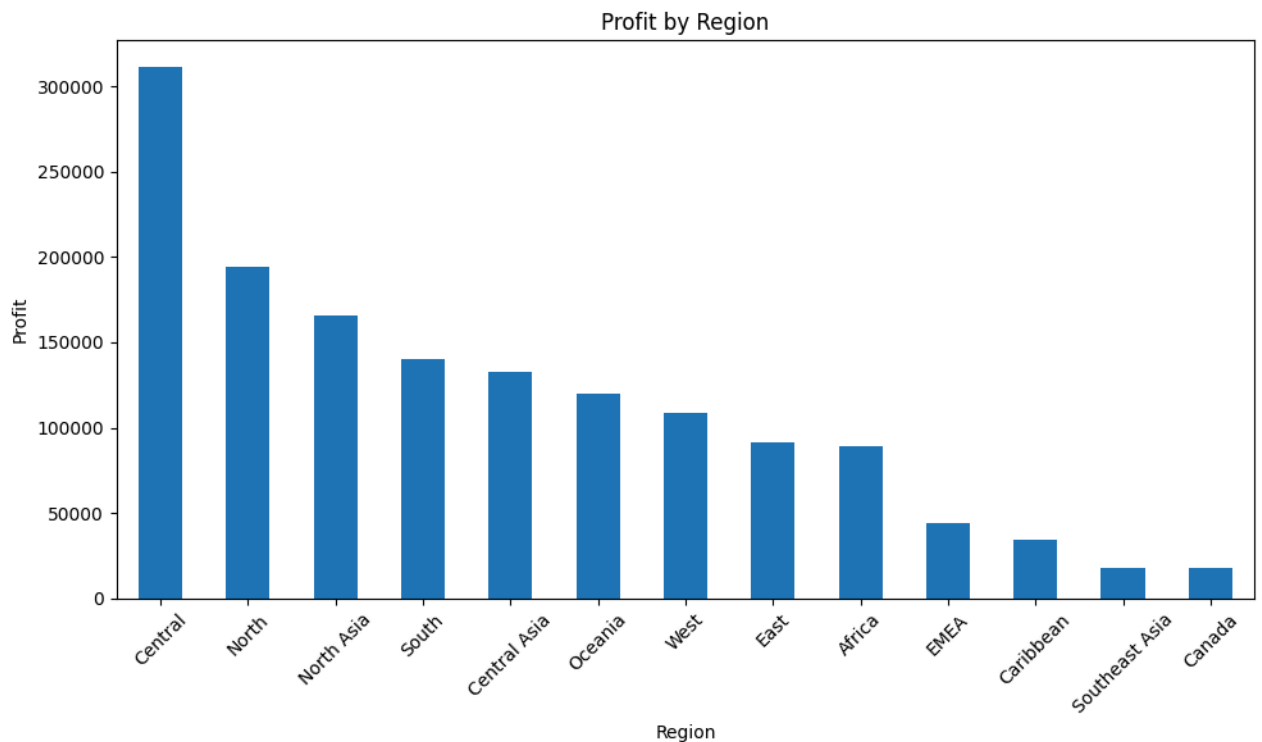
    plt.title("Profit by Region")
    plt.xlabel("Region")
    plt.ylabel("Profit")

    plt.xticks(rotation=45)

    plt.tight_layout()
```

```
plt.show()
```

```
=====
PROFIT BY REGION
=====
priyanshu sharma cu240250980
Region
Central      311403.98164
North        194597.95252
North Asia   165578.42100
South        140355.76618
Central Asia 132480.18700
Oceania       120089.11200
West         108418.44890
East          91522.78000
Africa        88871.63100
EMEA          43897.97100
Caribbean    34571.32104
Southeast Asia 17852.32900
Canada        17817.39000
Name: Profit, dtype: float64
```



```
# DATA STORY 6
# SALES BY SEGMENT
# =====

if "Segment" in df.columns and "Sales" in df.columns:

    segment_sales = (
        df.groupby("Segment")["Sales"]
        .sum()
        .sort_values(ascending=False)
    )

    print("\n=====")
    print("SALES BY CUSTOMER SEGMENT")
    print("=====")

    print(segment_sales)

    plt.figure(figsize=(10, 6))

    segment_sales.plot(
        kind="bar"
    )

    plt.title("Sales by Customer Segment")
    plt.xlabel("Customer Segment")
    plt.ylabel("Sales")
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
=====
```

```
SALES BY CUSTOMER SEGMENT
```

```
=====
```

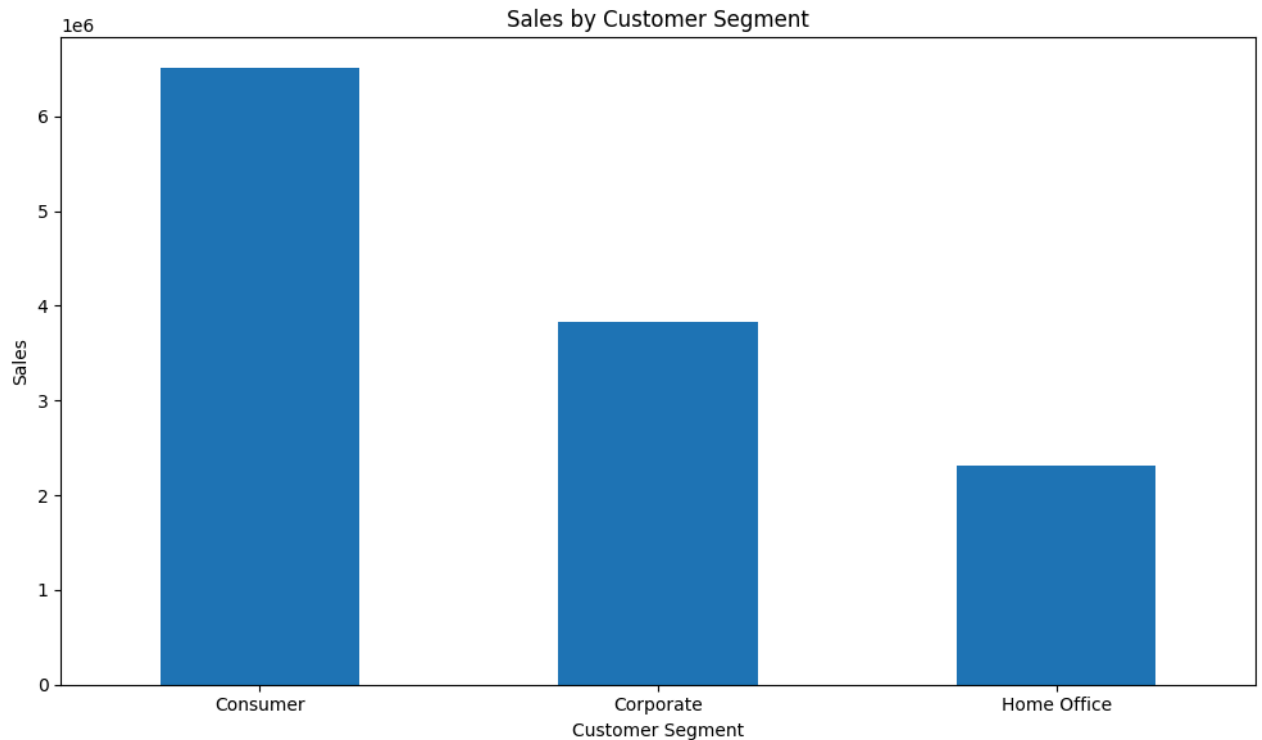
```
Segment
```

```
Consumer      6508141
```

```
Corporate      3824808
```

```
Home Office    2309956
```

```
Name: Sales, dtype: int64
```



```
# DATA STORY 7
```

```
# TOP 10 SUB-CATEGORIES
```

```
# =====
```

```
if "Sub-Category" in df.columns and "Sales" in df.columns:
```

```
    top_subcategories = (
        df.groupby("Sub-Category")["Sales"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
    )
```

```
    print("\n=====")
    print("TOP 10 SUB-CATEGORIES BY SALES")
    print("=====")
    print("priyanshu sharma cu240250980")
    print(top_subcategories)
```

```
    plt.figure(figsize=(12, 6))
```

```
    top_subcategories.sort_values().plot(
        kind="barh"
    )
```

```
    plt.title("Top 10 Sub-Categories by Sales")
    plt.xlabel("Sales")
    plt.ylabel("Sub-Category")
```

```
    plt.tight_layout()
    plt.show()
```

```
# DATA STORY 8
# TOP 10 PRODUCTS
# =====

if "Product Name" in df.columns and "Sales" in df.columns:

    top_products = (
        df.groupby("Product Name")["Sales"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
    )

    print("\n=====")
    print("TOP 10 PRODUCTS BY SALES")
    print("=====")

    print(top_products)
```

```
# =====
# DATA STORY 9
# CORRELATION HEATMAP
# =====

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Select numeric columns
numeric_columns = df.select_dtypes(
    include=np.number
).columns

print("\nNumeric Columns:")
print(numeric_columns.tolist())

if len(numeric_columns) > 1:

    # Calculate correlation
    correlation = df[numeric_columns].corr()

    print("\n=====")
    print("CORRELATION MATRIX")
    print("=====")

    print(correlation)

    # Create heatmap
    plt.figure(figsize=(10, 7))

    sns.heatmap(
        correlation,
        annot=True,
        cmap="coolwarm",
        fmt=".2f"
    )

    plt.title("Correlation Heatmap")

    plt.tight_layout()
    plt.show()

else:
    print("\nNot enough numeric columns for correlation analysis.")
```